

1. The number of unpaired electrons in Cr^+ will be

- (a) 3
- (b) 4
- (c) 5
- (d) 6

Answer: (c) 5

Explanation: $\text{Cr} = [\text{Ar}] 3d^5 4s^1$. Removing one 4s electron gives $\text{Cr}^+ = [\text{Ar}] 3d^5$, so it has 5 unpaired electrons.

2. The highest oxidation state of Cr will be

- (a) 2
- (b) 3
- (c) 4
- (d) 6

Answer: (d) 6

Explanation: Chromium reaches a maximum oxidation state of +6, for example in chromate/dichromate compounds.

3. Which statement is true about the transitional elements

- (a) They are highly reactive
- (b) They show variable oxidation states
- (c) They have low M.P.
- (d) They are highly electropositive

Answer: (b) They show variable oxidation states

Explanation: The correct choice is They show variable oxidation states. This

follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

4. The transitional metal which form green compound in +3 oxidation state and yellow orange compound in +6 oxidation state is

- (a) Fe
- (b) Ni
- (c) Cr
- (d) Co

Answer: (c) Cr

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

5. Highest (+7) oxidation state is shown by

- (a) Co
- (b) Cr
- (c) V
- (d) Mn

Answer: (d) Mn

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

6. Transitional elements are

- (a) All metals
- (b) Few metals and few non-metals
- (c) All solids



(d) All highly reactive

Answer: (a) All metals

Explanation: The correct choice is All metals. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

7. Which of the following has highest ionic radii

- (a) Cr+3
- (b) Mn+3
- (c) Fe+3
- (d) Co+3

Answer: (a) Cr+3

Explanation: The correct choice is Cr+3. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

8. In a reaction the ferrous (Fe⁺⁺) iron is oxidised to ferric (Fe⁺⁺⁺) ion. The equivalent weight of the ion in the above reaction is equal to

- (a) Half of the atomic weight
- (b) 1/5 of the atomic weight
- (c) The atomic weight
- (d) Twice the atomic weight

Answer: (c) The atomic weight

Explanation: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + e^-$. One electron is transferred, so equivalent weight = atomic weight/1 = atomic weight.

9. Which of the following element has maximum density

- (a) Hg
- (b) Au
- (c) Os
- (d) Pb

Answer: (c) Os

Explanation: Compare the relevant atomic masses and/or densities; the listed correct option has the greatest value among the choices.

10. Which is heaviest among the following

- (a) Iron
- (b) Copper
- (c) Gold
- (d) Silver

Answer: (c) Gold

Explanation: Compare the relevant atomic masses and/or densities; the listed correct option has the greatest value among the choices.

11. Transitional elements exhibit variable valencies because they release electrons from the following orbits

- (a) ns orbit
- (b) ns and np orbits
- (c) (n – 1)d and ns orbits
- (d) (n – 1)d orbit

Answer: (c) (n – 1)d and ns orbits



Explanation: The correct choice is (n – 1)d and ns orbits. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

12. The tendency towards complex formation is maximum in

- (a) s - block elements
- (b) p - block elements
- (c) d - block elements**
- (d) f - block elements

Answer: (c) d - block elements

Explanation: Transition-metal ions form complexes readily because of their relatively high charge density and available orbitals for accepting ligand electron pairs.

13. Which forms coloured

- (a) Metals
- (b) Non-metals
- (c) p - block elements
- (d) Transitional elements**

Answer: (d) Transitional elements

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

14. Which element belongs to d – block

- (a) Na
- (b) Ca
- (c) Cu**
- (d) Ar

Answer: (c) Cu

Explanation: The correct choice is Cu. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

15. Variable valency is shown by

- (a) Na
- (b) Cu**
- (c) Mg
- (d) Al

Answer: (b) Cu

Explanation: Transition elements can use both ns and (n-1)d electrons in bonding because these levels are close in energy, so several oxidation states are possible.

16. The element with a atomic number 26 is

- (a) A non-metal
- (b) Krypton
- (c) Iron**
- (d) Manganese

Answer: (c) Iron

Explanation: The correct choice is Iron. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.



17. One of the following metals forms a volatile carbonyl compound and this property is taken advantage of for its extraction. This metal is

- (a) Iron
- (b) Nickel**
- (c) Cobalt
- (d) Tungston

Answer: (b) Nickel

Explanation: Nickel forms volatile Ni(CO)_4 in the Mond process, which is used for purification/extraction of nickel.

18. The coinage metals are

- (a) Iron, Cobalt, Nickel
- (b) Copper and Zinc
- (c) Copper, Silver and Gold**
- (d) Gold and Platinum

Answer: (c) Copper, Silver and Gold

Explanation: The traditional coinage metals are Cu, Ag and Au.

19. Which of the following structure is that of a coinage metal

- (a) 2, 8, 1
- (b) 2, 8, 18, 1**
- (c) 2, 8, 8
- (d) 2, 18, 8, 3

Answer: (b) 2, 8, 18, 1

Explanation: The correct choice is 2, 8, 18, 1. This follows from the standard

periodic and electronic properties of d- and f-block elements tested in this question.

20. An elements in +3 oxidation state has the electronic configuration $(\text{Ar})3d^3$. Its atomic number is

- (a) 24**
- (b) 23
- (c) 22
- (d) 21

Answer: (a) 24

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before $(n-1)d$ electrons.

